Introduction to

Machine Learning

with Orange Data Mining

Version 1.0.0

Chapter Supervised Learning	Chapter Unsupervised Learning	Chapter Time Series Analysis
Classification & Regression	Clustering	Pattern Over Time

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Introduction

What is Machine Learning?

Machine learning is a way of teaching computers to make decisions or predictions by showing them examples, rather than writing out every rule manually.

Think about how you learned to recognize a cat. Nobody gave you a rulebook. You simply saw many cats over time and your brain learned the pattern. Machine learning works the same way.

> **Core Idea**
>
> Machine Learning = Teaching a computer to find patterns in data so it can make predictions on data it has never seen before.
>
> You provide: Examples (data)
> The computer finds: Patterns (a model)
> The result: Predictions on new, unseen data

The Four Topics in This Handbook

Topic	Type	Question It Answers
Classification	Supervised	Which category does this belong to?
Regression	Supervised	What number value will this be?
Clustering	Unsupervised	What natural groups exist in this data?
Time Series	Pattern over time	What will happen next based on the past?

Supervised vs Unsupervised

Supervised Learning	Unsupervised Learning
You provide labelled data — you tell it the correct answer	You provide unlabelled data — no answers given
The model learns to predict the label	The model finds hidden structure on its own
Example: Email is spam or not spam	Example: Group customers by behaviour
Covered in Chapters 1 and 2	Covered in Chapter 3

Getting Started with Orange

Orange is a free, open-source visual tool for machine learning. Instead of writing code, you connect boxes (widgets) together to build a workflow. Every step is visible, making it easy to understand what is happening.

Installing Orange

1. Visit https://orangedatamining.com
2. Click Download and install for your operating system
3. Open Orange — you will see a blank canvas
4. Widgets are listed in the left panel
5. Drag widgets onto the canvas and connect them by drawing lines between their ports

How to Use This Handbook

Read the theory sections to understand the concept.
Follow the hands-on steps in Orange — in order, one step at a time.
Check your progress with the Checkpoint boxes.
Complete the assignment at the end of each chapter.

Classification

1.1 What is Classification?

Classification is the task of sorting data into categories. A model looks at the features of an input and assigns it to one of a fixed set of categories (called classes).

Everyday Examples

Email filtering: Is this message spam or not spam?
Medicine: Does this patient have the disease or not?
Banking: Is this transaction fraudulent or legitimate?
Botany: Is this flower a rose, tulip, or daisy?

1.2 Key Vocabulary

Term	What It Means
Dataset	A table of data. Rows are examples. Columns are attributes.
Feature	An input column — information used to make a prediction
Target / Label	The output column — the answer the model tries to predict
Instance	One row in the dataset — one example
Model	The pattern an algorithm learned from training data
Class	One possible category in the target column

1.3 The ML Pipeline

Every machine learning task follows the same sequence. Learn this — it applies to every chapter in this handbook.

The 5-Step ML Pipeline

Step 1: Load your data
Step 2: Explore and understand your data
Step 3: Split data into training set and test set
Step 4: Train a model on the training set
Step 5: Evaluate the model on the test set

1.4 Hands-On: Loading and Exploring Data

The Dataset — Iris

The Iris dataset contains measurements of 150 flowers from three species. It is one of the most widely used introductory datasets in machine learning.

Column	Role
sepal length	Feature — length of the outer petal (cm)
sepal width	Feature — width of the outer petal (cm)
petal length	Feature — length of the inner petal (cm)
petal width	Feature — width of the inner petal (cm)
iris	TARGET — the species we want to predict

Step 1 — Load the Data

Orange Instructions

1. Open Orange
2. In the left panel, find the Data category
3. Drag the 'Datasets' widget onto the canvas
4. Double-click it to open, then search for 'iris'
5. Select the Iris dataset

Checkpoint 1.1

Datasets widget shows: 150 instances, 5 features
The widget is active on the canvas (no red border)

Step 2 — Explore with Data Table

Orange Instructions

1. Drag a 'Data Table' widget onto the canvas
2. Connect: Datasets → Data Table
3. Double-click Data Table to open it
4. Observe: each row is one flower; the last column is the species (target)

Step 3 — Explore with Distributions

Orange Instructions

1. Drag a 'Distributions' widget onto the canvas
2. Connect: Datasets → Distributions
3. Use the Variable dropdown to switch between columns
4. Observe how values are spread for each feature

What to Look For

Are the three species mixed together or separated in each feature?
Which features seem most useful for telling species apart?
Are there any extreme values (outliers)?

Step 4 — Scatter Plot

Orange Instructions

1. Drag a 'Scatter Plot' widget onto the canvas
2. Connect: Datasets → Scatter Plot
3. Set X-axis to 'petal length', Y-axis to 'petal width'
4. Set Color to 'iris'

Checkpoint 1.2

Three distinct colored groups appear
One group (Iris setosa) is clearly separated from the other two
This tells us petal measurements are strong predictors of species

1.5 Hands-On: Train / Test Split

You need to test your model on data it has never seen. If you train and test on the same data, the results will be misleadingly good — the model has simply memorized the answers.

The Exam Analogy

Training set (~80%): Study material — the model learns from this
Test set (~20%): Exam questions — used only to evaluate the model

Never test on training data. Results will be artificially perfect.

Orange Instructions — Data Sampler

1. Drag a 'Data Sampler' widget onto the canvas
2. Connect: Datasets → Data Sampler
3. Set sampling to 'Fixed proportion of data' at 80%
4. Check 'Replicable (fixed random seed)'
5. Click 'Sample Data'
   Top output = training set (80%)
   Bottom output = test set (20%)

1.6 Hands-On: Decision Tree

How It Works

A decision tree makes predictions by asking a series of yes/no questions in sequence — like a flowchart. It automatically learns which questions to ask and in what order.

Simple Analogy

Is the petal width <= 0.8 cm?
  YES → Iris setosa
  NO → Is the petal length <= 4.9 cm?
    YES → Iris versicolor
    NO → Iris virginica

The model learned these thresholds from the training data.

Orange Instructions — Build the Workflow

1. Drag a 'Tree' widget onto the canvas
2. Drag a 'Test and Score' widget onto the canvas
3. Connect: Data Sampler (train) → Tree
4. Connect: Data Sampler (train) → Test and Score
5. Connect: Data Sampler (test) → Test and Score (second input)
6. Connect: Tree → Test and Score

Orange Instructions — Tree Viewer

1. Drag a 'Tree Viewer' widget onto the canvas
2. Connect: Tree → Tree Viewer
3. Open Tree Viewer to see the flowchart your model learned
4. Each node shows the question asked and each leaf shows the predicted class

Checkpoint 1.3

Tree Viewer shows a branching flowchart
Each internal node contains a condition (e.g. petal width <= 0.8)
Each leaf node shows one predicted class

1.7 Hands-On: Evaluating Results

Confusion Matrix

The confusion matrix shows exactly where your model was correct and where it was wrong. Read it before looking at any other metric.

Orange Instructions

1. Drag a 'Confusion Matrix' widget
2. Connect: Test and Score → Confusion Matrix

Term	What It Means
True Positive (TP)	Predicted class A — actually was class A. Correct.
True Negative (TN)	Predicted NOT class A — really was not. Correct.
False Positive (FP)	Predicted class A — was actually something else. Wrong.
False Negative (FN)	Predicted NOT class A — actually was class A. Wrong.

Accuracy

The simplest metric. Out of all predictions, how many were correct?

```
Accuracy = (TP + TN) / Total Predictions
```

When Accuracy Misleads

If 95 out of 100 patients are healthy, a model always predicts 'healthy' achieves 95% accuracy but it never detects a sick patient. Always inspect the confusion matrix alongside accuracy.

Precision and Recall

Metric	Question It Answers
Precision	When I predicted positive, how often was I actually right?
Recall	Of all the actual positives, how many did I successfully catch?

```
Precision = TP / (TP + FP)        Recall = TP / (TP + FN)
```

When Each Metric Matters

High Precision: False alarms are costly (e.g. spam filter blocking real emails)
High Recall: Missing a case is costly (e.g. medical test missing sick patients)
There is a natural trade-off — improving one often reduces the other.

F1 Score

F1 balances precision and recall into a single number. Most useful when classes are imbalanced.

```
F1 = 2 x (Precision x Recall) / (Precision + Recall)
```

F1 Value	Interpretation
Close to 1.0	Good balance of precision and recall
Close to 0.0	Poor performance on at least one dimension

ROC Curve and AUC

The ROC curve shows how well the model separates classes at different decision thresholds. AUC summarizes the entire curve into one number.

Orange Instructions

1. Drag a 'ROC Analysis' widget
2. Connect: Test and Score → ROC Analysis, a curve closer to the top-left corner is better

AUC	Interpretation
1.0	Perfect
0.9 – 1.0	Excellent
0.7 – 0.9	Good
0.5	Random guessing — no better than a coin flip
< 0.5	Worse than random — check your workflow

Metrics at a Glance

Metric	What It Asks	Use When
Accuracy	How often was I correct overall?	Classes are balanced
Precision	When I said positive, was I right?	False alarms are costly
Recall	Did I catch all real positives?	Missing a case is costly
F1 Score	Do I balance precision and recall?	Classes are imbalanced
AUC	Is the model better than guessing?	Comparing models overall

1.8 Hands-On: Comparing Models

Orange makes model comparison easy — connect multiple learner widgets to one Test and Score widget and compare them side by side.

Model	How It Works
Decision Tree	Asks yes/no questions in sequence — like a flowchart
k-Nearest Neighbours	Finds the k most similar examples and takes a vote
Naive Bayes	Uses probability; assumes features are independent
Logistic Regression	Finds a boundary that separates classes

Orange Instructions

1. Add 'kNN', 'Naive Bayes', and 'Logistic Regression' widgets
2. Connect each to the same Test and Score widget
3. Open Test and Score — one row per model
4. Compare AUC, CA (Classification Accuracy), F1, Precision, Recall

Checkpoint 1.4

Test and Score show one row per model
Results differ between models — same data, different algorithms
Note: the 'best' model depends on which metric matters for your goal

1.9 Recap

You have completed a full classification pipeline:

1. Loaded and explored the Iris dataset
2. Split data into training and test sets
3. Trained a Decision Tree and read its logic visually
4. Evaluated with Confusion Matrix, ROC, and Test and Score
5. Compared four models side by side

Common Mistakes to Avoid

Testing on training data — always split first
Relying on accuracy alone — always check the confusion matrix
Ignoring the confusion matrix — it shows WHERE the model fails
Assuming one model is always best — it depends on your metric and goal

1.10 Assignment

Reflection Questions

6. In your own words, what is the difference between a feature and a target?
7. Why must the test set never be seen by the model during training?
8. A model has 99% accuracy. Does that guarantee it is a good model? Explain.
9. Give a real-world scenario where recall matters more than precision.
10. In your confusion matrix, which class did the model struggle with most? Why?

Hands-On Exercise

Exercise 1 — Titanic Dataset

1. Load the 'Titanic' dataset from the Datasets widget
   Target column: 'survived'
2. Build the complete pipeline: load → explore → split → train → evaluate
3. Train all four models: Tree, kNN, Naive Bayes, Logistic Regression
4. Answer in writing:
   a. Which model had the highest AUC?
   b. Which model had the highest F1?
   c. Are these the same model? Which would you choose and why?
   d. How many passengers did the best model incorrectly predict as 'survived' when they did not? (Read from the Confusion Matrix)
5. Submit a screenshot of your Test and Score results

Regression

2.1 What is Regression?

Regression is like classification — it uses features to make predictions. The difference: instead of predicting a category, regression predicts a number.

Everyday Examples

House price: Given size, location, rooms — predict price in dollars
Temperature: Given past weather patterns — predict tomorrow's temperature
Student grade: Given study hours — predict exam score
Sales forecast: Given advertising spend — predict next month's revenue

2.2 Classification vs Regression

Classification	Regression
Predicts a category	Predicts a number
Output: 'spam' or 'not spam'	Output: '$245,000'
Metrics: Accuracy, F1, AUC	Metrics: MAE, RMSE, R^2
Example: Is this tumor malignant?	Example: How large is this tumor?

2.3 Key Vocabulary

Term	What It Means
Continuous value	A number that can take any value (e.g. 3.7, 150.2)
Residual / Error	The difference between the predicted value and actual value
MAE	Mean Absolute Error — average size of errors
RMSE	Root Mean Squared Error — penalizes large errors more
R^2	How much of the variation in the data the model explains (0 to 1)

2.4 Hands-On: Loading and Exploring Data

The Dataset — Housing

The Housing dataset contains information about houses and their prices. The goal is to predict house price based on characteristics of the property and neighbourhoods.

Orange Instructions

1. Open the Datasets widget
2. Search for 'housing' and select it
3. Connect to Data Table — find the target column 'MEDV' (median house value)
4. Connect to Distributions — explore the MEDV column
   Question: What does the distribution of house prices look like?

Checkpoint 2.1

506 instances, 14 columns
The target column MEDV contains numbers, not categories
Most prices cluster in a range, with some high outliers

2.5 Hands-On: Building a Regression Model

Orange Instructions — Full Workflow

1. Connect: Datasets → Data Sampler (80% train / 20% test)
2. Drag a 'Linear Regression' widget
3. Connect: Data Sampler (train) → Linear Regression
4. Drag a 'Test and Score' widget
5. Connect: Data Sampler (train) → Test and Score
6. Connect: Data Sampler (test) → Test and Score (second input)
7. Connect: Linear Regression → Test and Score
8. Open Test and Score

2.6 Reading Regression Metrics

MAE — Mean Absolute Error

MAE is the average size of your prediction errors, regardless of direction. Simple to interpret: if MAE = 3.5, your predictions are off by 3.5 units on average.

```
MAE = average of | predicted - actual |
```

RMSE — Root Mean Squared Error

RMSE is similar to MAE but punishes large errors disproportionately. A prediction off by 10 contributes far more than ten predictions off by 1.

```
RMSE = square root of (average of (predicted - actual) ²)
```

Use RMSE when large errors are especially harmful — medical dosage, structural loads, financial risk.

R^2 — Coefficient of Determination

R^2 tells you how much of the variation in the target your model explains. Think of it as the percentage of the pattern the model captured.

R^2 Value	Interpretation
1.0	Perfect — model explains all variation
0.8 – 1.0	Very good
0.5 – 0.8	Moderate — captures some but not all patterns
Below 0.5	Weak — model misses significant structure
0 or negative	Worse than simply predicting the mean value

2.7 Hands-On: Predictions vs Actual

Orange Instructions

1. Drag a 'Predictions' widget
2. Connect: Linear Regression + Data Sampler (test) → Predictions
3. Open Predictions
4. Each row shows the actual value and the model's prediction side by side
5. Identify the rows with the largest gaps — these are the hardest cases

Checkpoint 2.2

Predictions widget shows actual and predicted values per row
Most predictions are close to actual values
A few rows show larger gaps — note which direction (over or under predicted)

2.8 Hands-On: Comparing Models

Orange Instructions

1. Add 'kNN' and 'Random Forest' widgets
2. Connect both to the same Test and Score widget as Linear Regression
3. Compare MAE, RMSE, and R^2 across all three models

What is Random Forest?

Random Forest builds many decision trees and combines their predictions.
It often outperforms a single tree because it reduces overfitting.
Overfitting = model memorized training data and fails on new data.

2.9 Recap

Classification (Ch. 1)	Regression (Ch. 2)
Predicts categories	Predicts numbers
Iris species, spam/not spam	House price, temperature
Accuracy, F1, AUC	MAE, RMSE, R^2
Confusion Matrix	Predictions table
Tree, kNN, Naive Bayes, Logistic	Linear Regression, kNN, Random Forest

Common Mistakes to Avoid

Using classification metrics for regression — they measure different things
Low MAE but high RMSE — means there are occasional large errors
High R^2 but large MAE — model explains patterns but errors are still big

2.10 Assignment

Reflection Questions

11. What is the fundamental difference between classification and regression?
12. If MAE = 2.0 and RMSE = 9.5, what does the gap tell you about the error distribution?
13. $R^2 = 0.45$ — is this a good model? What would your next step be?
14. Why might Random Forest outperform a single Decision Tree?
15. You need to predict a patient's blood pressure. Is this classification or regression? Explain.

Hands-On Exercise

Exercise 2 — Predict a Continuous Value

1. Load the 'Housing' dataset (or 'Auto MPG' from Datasets)
2. Identify the target column
3. Build the full regression pipeline
4. Train: Linear Regression, kNN, Random Forest
5. Answer in writing:
   a. Which model had the lowest MAE? By how much?
   b. Which model had the best R^2?
   c. Open Predictions for your best model.
      Find the row with the largest error. Predicted vs actual?
   d. What does this suggest about what the model struggles with?
6. Submit a screenshot of your Test and Score results

Clustering

3.1 What is Clustering?

Clustering is the task of grouping similar data points together — without being told what the groups should be. The algorithm discovers structure in the data on its own.

The fundamental difference from classification: there is no target column. You do not tell the model the correct answer. You let it find natural groupings.

Everyday Examples

Retail: Group customers by purchasing behaviours
Biology: Group genes by expression patterns to find gene families
News: Automatically group articles by topic
Medicine: Group patients by symptoms to identify unknown conditions

3.2 Supervised vs Unsupervised

Supervised (Ch. 1–2)	Unsupervised (Ch. 3)
Has labelled data (target column)	No labels — no target column
Model learns from correct answers	Model finds structure independently
Evaluated with Accuracy, F1, MAE	Evaluation is harder — no right answer
Classification and Regression	Clustering

3.3 How k-Means Works

k-Means is the most widely used clustering algorithm. It divides data into k groups, where k is a number you choose.

k-Means Algorithm

Step 1: Choose k (the number of clusters you want to find)
Step 2: Place k random center points (centroids) in the data space
Step 3: Assign every data point to its nearest centroid
Step 4: Move each centroid to the center of its assigned points
Step 5: Repeat steps 3 and 4 until centroids stop moving
Result: k groups of similar data points

What Is k?

k is the number of groups you ask the algorithm to find.
k = 3 means: find 3 groups in this data.
Choosing k is one of the key decisions — covered in Section 3.6.

3.4 Hands-On: Clustering the Iris Dataset

We will reuse the Iris dataset — but this time we pretend we do not know the species labels. We will see whether k-Means can discover the three species on its own.

Orange Instructions

1. Load the Iris dataset using Datasets widget
2. Drag a 'k-Means' widget onto the canvas
3. Connect: Datasets → k-Means
4. Open k-Means — set Number of Clusters to 3
5. Click Apply

Checkpoint 3.1

k-Means output adds a new column called 'Cluster'
Each row is assigned to Cluster C1, C2, or C3
Connect to Data Table to see the assignments

3.5 Hands-On: Visualizing Clusters

Scatter Plot

Orange Instructions

1. Connect: k-Means → Scatter Plot
2. Set X-axis: petal length, Y-axis: petal width
3. Set Color: Cluster
4. Compare to your Chapter 1 scatter plot (colored by actual species)
   Question: How well do the clusters match the actual species?

Silhouette Plot

The silhouette plot evaluates how well each data point fits within its assigned cluster versus neighbouring clusters. Higher scores mean cleaner separation.

Orange Instructions

1. Drag a 'Silhouette Plot' widget
2. Connect: k-Means → Silhouette Plot
3. Open it and examine the distribution of scores

Silhouette Score	Interpretation
Close to 1.0	Point fits well — clearly belongs in its cluster
Around 0	Point is on the boundary between two clusters
Negative	Point may belong in a different cluster

3.6 Choosing k

There is no universally correct value for k. Too few clusters miss structure; too many over-splits natural groups. The most practical approach: try several values and compare.

Orange Instructions — Compare k Values

1. Open the k-Means widget
2. Set k = 2, note the average silhouette score
3. Set k = 3, note the average silhouette score
4. Set k = 4, note the average silhouette score
5. Choose the k that gives the highest average silhouette score and produces clusters that make practical sense

Rule of Thumb

Look for the k that maximizes average silhouette score.
Prefer fewer clusters when scores are similar — simpler is better.
Always ask: do these clusters make sense in the real world?

3.7 Hands-On: Interpreting Clusters

Orange labels clusters C1, C2, C3. It does not name them. That is your job as the analyst — look at what the points in each cluster have in common.

Orange Instructions

1. Connect: k-Means output → Data Table
2. Sort by the Cluster column
3. Examine feature values for each cluster
4. Ask: what do points in Cluster 1 share? What about Cluster 2?

What to Look For

Are values in one cluster consistently higher or lower for any feature?
Do the clusters correspond to something meaningful in context?
Give each cluster a descriptive name based on your observations.

3.8 Recap

16. Loaded unlabelled data — no target column
17. Applied k-Means to find natural groupings
18. Visualized clusters with Scatter Plot and Silhouette Plot
19. Compared k values using silhouette scores
20. Interpreted and named clusters from feature patterns

Common Mistakes to Avoid

Treating clusters as ground truth — they are patterns, not facts
Defaulting to k = 2 — more clusters may reveal richer structure
Skipping visualization — numbers alone are never enough
Forgetting context — a good cluster must be meaningful, not just mathematically clean

3.9 Assignment

Reflection Questions

21. In your own words, what is the difference between classification and clustering?
22. Why is there no accuracy score in clustering?
23. A data point has a silhouette score of -0.3. What does that mean?
24. You test k = 2, 3, 4. k = 3 gives the best silhouette score. What would you choose and why?
25. Could you use clustering as a step before classification? How might that help?

Hands-On Exercise

Exercise 3 — Zoo Dataset

1. Load the 'Zoo' dataset from the Datasets widget
   (101 animals, each described by biological features)
2. Use the Select Columns widget to remove 'name' and 'type'
   (So, k-Means does not use the labels)
3. Apply k-Means with k = 3, then k = 5, then k = 7
4. For each k, record the average silhouette score
5. Answer in writing:
   a. Which k gave the best silhouette score?
   b. Open Data Table sorted by cluster for your best k.
      What do the animals in each cluster have in common?
   c. Give each cluster a descriptive name
6. Submit a screenshot of your Scatter Plot and Silhouette Plot

Time Series Analysis

4.1 What is Time Series Data?

Time series data is any data recorded at regular intervals over time. The critical distinction: the order of rows matters. What happened at time T-1 influences what happens at time T.

Everyday Examples

Stock prices: Recorded every second, minute, or day
Weather: Temperature measured every hour
Sales: Monthly revenue over several years

4.2 What Makes Time Data Different?

Regular Data	Time Series Data
Row order does not matter	Row order is critical — time flows forward
Each row is independent	Each value depends on prior values
Can shuffle rows freely	Shuffling destroys meaning
Train/test split is random	Split must respect chronological order
Features are measurements	Features include past values (lags)

4.3 Key Concepts

Trend

A long-term increase or decrease in the data over time. Example: global average temperatures rising steadily over decades.

Seasonality

A repeating pattern tied to a calendar cycle. Example: ice cream sales peak every summer and drop every winter — predictably.

Noise

Random variation that cannot be explained by trend or seasonality. Example: an unexpected viral post causing a one-day spike in web traffic.

Time Series = Trend + Seasonality + Noise

Most time series contain all three components.
A good model identifies and learns from the first two.
Noise is random, it cannot be predicted.

Lag

A lag is a past value of the variable used as a feature. 'Yesterday's temperature' is a lag-1 feature for predicting today's temperature. Lag features allow the model to learn from its own history.

4.4 Hands-On: Setting Up Orange for Time Series

Orange requires a separate Time Series add-on. Install it before proceeding.

Install the Add-On

1. In Orange, go to Options → Add-ons
2. Find 'Time Series' in the list
3. Check the box and click OK
4. Restart Orange

4.5 Hands-On: Loading and Visualizing

Orange Instructions — Load Data

1. Find the Time Series category in the widget panel
2. Drag the 'Yahoo Finance' widget onto the canvas
   Enter a stock symbol (e.g. AAPL) and set a date range (2 years)
3. Connect to a 'Line Chart' widget
4. Open Line Chart

Checkpoint 4.1

Line Chart shows data plotted over time (time on the X axis)
You can visually identify whether there is an upward or downward trend
You can see whether there is a repeating wave pattern (seasonality)

Periodogram

The periodogram identifies dominant repeating cycles in your data. A strong peak at period = 12 means the data repeats every 12-time steps.

Orange Instructions

1. Connect your time series data → Periodogram widget
2. Open Periodogram
3. The highest peaks indicate the dominant cycles

4.6 Hands-On: Building a Forecast Model

The VAR (Vector AutoRegression) model predicts future values based on patterns in past values.

Orange Instructions

1. Connect your time series data → 'VAR' model widget
2. Open it — set the number of lags (how many past steps to use)
3. Connect: VAR → 'Forecast' widget
4. Set how many future steps to predict (e.g. 30 days)
5. Connect: Forecast → Line Chart

Checkpoint 4.2

Line Chart shows historical data in one color
Forecast appears as an extension of the line
A shaded band shows prediction uncertainty — it widens over time

4.7 Evaluating the Forecast

Forecasting predicts numbers, so we use the same metrics as regression:

Metric	What It Measures	Notes
MAE	Average size of forecast errors	Easy to interpret in original units
RMSE	Error weighted by magnitude	Penalizes large misses more than MAE
MAPE (%)	Error as a percentage of actual value	Useful for comparing across scales

> **Critical: Time-Aware Train/Test Split**
> Never randomly split time series data.

4.8 Recap

26. Understood what makes time series data structurally different
27. Identified trend, seasonality, and noise in a real dataset
28. Visualized time series using Line Chart and Periodogram
29. Built a forecasting model in Orange using VAR
30. Evaluated forecast quality with MAE, RMSE, and MAPE

> **Common Mistakes to Avoid**
> Ignoring trend — a growing series needs a model that accounts for growth
> Forecasting too far ahead — uncertainty grows rapidly; short forecasts are more reliable
> Treating noise as signal — some variation is purely random and cannot be modelled

4.9 Assignment

Reflection Questions

31. What are the three components of a time series? Give a real example of each.
32. Why is it wrong to randomly split time series data for train/test evaluation?
33. MAE = 5.0 but MAPE = 45%. What does this combination suggest about the data?
34. You see a strong peak at period = 7 in the Periodogram. What does that mean?
35. When would forecasting be more valuable than classification? Give one scenario.

Hands-On Exercise

> **Exercise 4 — Stock Price Forecast**
> 1. Use the Yahoo Finance widget to load 2 years of daily data
> for any stock of your choice (e.g. AAPL, TSLA, GOOGL)
> 2. Open Line Chart — describe the trend you observe
> 3. Open Periodogram — identify the dominant cycle
> 4. Build a forecast for the next 30 days using VAR
> 5. Answer in writing:
> a. What trend did you observe in the historical data?
> b. What does the model predict for the next 30 days?
> c. How confident are you in this forecast, and why?
> 6. Submit a screenshot of your Line Chart with the forecast visible

Appendix

A. Orange Widget Reference

Widget	Category	Purpose
Datasets	Data	Load built-in datasets (Iris, Housing, etc.)
Data Table	Data	View raw data in rows and columns
Select Columns	Data	Include or exclude specific columns
Distributions	Visualize	Distribution of values per column
Scatter Plot	Visualize	Plot two features against each other
Data Sampler	Data	Split data into train and test sets
Tree	Model	Train a Decision Tree
Tree Viewer	Visualize	Visual flowchart of the Decision Tree
kNN	Model	k-Nearest Neighbors
Naive Bayes	Model	Probabilistic classifier
Logistic Regression	Model	Linear boundary classifier
Linear Regression	Model	Predict continuous values linearly
Random Forest	Model	Ensemble of Decision Trees
Test and Score	Evaluate	Compare multiple models on metrics
Confusion Matrix	Evaluate	Classification error breakdown
ROC Analysis	Evaluate	ROC curves and AUC scores
Predictions	Evaluate	Predicted vs actual values per row
k-Means	Cluster	k-Means clustering algorithm
Silhouette Plot	Visualize	Cluster quality evaluation
Line Chart	Time Series	Visualize time series data
Periodogram	Time Series	Identify dominant cycles
VAR / ARIMA	Time Series	Forecast model
Forecast	Time Series	Generate future predictions

B. Metrics Reference Card

Classification

Metric	Formula	Use When
Accuracy	(TP+TN) / Total	Classes are balanced
Precision	TP / (TP+FP)	False alarms are costly
Recall	TP / (TP+FN)	Missing a case is costly
F1 Score	2×(P×R)/(P+R)	Classes are imbalanced
AUC	Area under ROC	Comparing models overall

Regression

Metric	Interpretation	Use When
MAE	Average error in original units	All errors matter equally
RMSE	Error weighted by size	Large errors are especially costly
R^2	Proportion of variance explained (0–1)	Comparing models on same dataset

Clustering

Metric	Interpretation
Silhouette Score (per point)	How well a point fits its cluster (-1 to 1, higher is better)
Average Silhouette Score	Overall clustering quality — use to compare values of k

Time Series

Metric	Interpretation
MAE	Average forecast error in original units
RMSE	Forecast error, penalizing large misses more heavily
MAPE (%)	Error as a percentage — good for cross-scale comparison

C. Glossary

Term	Definition
Algorithm	A set of rules a computer follows to learn patterns from data
AUC	Area Under the ROC Curve — summarizes model performance in one number
Class	One possible category in a classification target
Cluster	A group of similar data points identified by an unsupervised algorithm
Confusion Matrix	A table showing correct and incorrect predictions in detail
Continuous value	A number that can take any real value (e.g. 3.7, 150.2)
Dataset	A structured table of data used for training or testing a model
Feature	An input column used to make a prediction
F1 Score	A metric that balances Precision and Recall into one number
Forecast	A prediction of future values in a time series
Instance	One row in a dataset — one example
k (k-Means)	The number of clusters you instruct the algorithm to find
k (k-NN)	The number of nearest neighbors the algorithm consults
Label	The known correct answer for a training example — same as Target
Lag	A past value of a variable used as a feature in time series models
MAE	Mean Absolute Error — average size of prediction errors
Model	The pattern an algorithm learned from training data
Noise	Random variation in data that cannot be modeled or predicted
Overfitting	When a model memorizes training data and fails to generalize
Periodogram	A chart identifying the strength of repeating cycles in time series data
Precision	Of all positive predictions, the fraction that were actually positive
R^2	How much of the target's variation the model explains (0 to 1)

Recall	Of all actual positives, the fraction the model correctly identified
Regression	Predicting a continuous numerical value
RMSE	Root Mean Squared Error — penalizes large errors more than MAE
ROC Curve	A graph of model performance across different decision thresholds
Seasonality	A repeating pattern in time series data tied to a calendar cycle
Silhouette Score	A measure of how well a data point fits its assigned cluster
Supervised Learning	Learning from labeled data — both features and target provided
Target	The output column — what the model is trained to predict
Test Set	Data held back from training, used only to evaluate the model
Training Set	Data the model learns from
Trend	A long-term directional movement in a time series
Unsupervised Learning	Learning from unlabeled data — finding hidden structure

D. Dataset Reference

Dataset	Chapter	Where to Find It
Iris	Chapter 1 — Classification	Orange: Datasets widget → search 'iris'
Titanic	Chapter 1 — Assignment	Orange: Datasets widget → search 'titanic'
Housing	Chapter 2 — Regression	Orange: Datasets widget → search 'housing'
Zoo	Chapter 3 — Assignment	Orange: Datasets widget → search 'zoo'
Stock data (any)	Chapter 4 — Time Series	Orange: Time Series → Yahoo Finance widget